MATH1014 Algebra

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Contents

Preface

THESE NOTES follow, and are meant to cement the course *Algebra*, taught at the University of Western Australia by Dr. Gordon Royle in Semester 2 of 2026. Where needed, the content of lectures is supplemented by *Linear Algebra Done Right* by Sheldon Axler.

0.1 What is Algebra?

I DO NOT YET KNOW ENOUGH TO SAY.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every *or* for all
$\exists$	There exists
$\in$	In *or* is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not *or* without
$\varnothing$	Empty set
$\sim$	Relates to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$\lvert n\rvert$	The absolute value of n
$\square$	It has been proven

Chapter 1

Fields and Rings

1.1 Lecture 1

1.1.1 Binary Operations

BINARY OPERATIONS are *operations*[1] that take two inputs and produce one output. Addition (+) and multiplication (×) are common examples of binary operations.

A uniary operation is one that takes only one input, and produces one output; for example, sine. The function $f(x, y, z) = x + y + z$ is an example of a *ternary* operation. There are an infinite amount of n-ary operations, however binary operations form the basis of the study of fields and rings.

1.1.2 Axioms

AXIOMS are statements, assumed to be true, used to define the domain of a particular mathematical theory. The goal is the find the smallest set of obviously true statements that define the domain which we are interested. If any axioms are logical consequences of others, they should be omitted, and likewise if any properties of the underlying structure we wish to define are lost, an axiom should be added to its definition to provide those properties. Axioms form a basis for further reasoning. We cannot prove anything without first agreeing on the postulates of our theories.

1.1.3 The Definition of a Field

A FIELD is a *set* S, equipped with the two binary operations + and ×, that behave similarly to how addition and multiplication would behave on real numbers, that also satisfies the following *axioms*:

[1] Operations are not yet defined.

1. Commutativity of +. $[a+b=b+a]$
2. Associativity of +. $[a+(b+c)=(a+b)+c]$
3. Existence of the additive identity 0. $[a+0=a]$
4. Existence of the additive inverse. $[\forall a \in S,\ \exists b \in S : a+b=0]$
5. Commutativity of ×. $[a \times b = b \times a]$
6. Associativity of ×. $[a \times (b \times c) = (a \times b) \times c]$
7. Existence of the multiplicative identity 1. $[a \times 1 = a]$
8. Existence of the multiplicative inverse. $[\forall a \in S,\ \exists b \in S : a \times b = 1]$
9. Distributivity. $[a \times (b+c) = a \times b + a \times c]$
10. Non-triviality. $[0 \neq 1]$

1.1.4 $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Q}$ are Fields

Starting with the reals, it is quite clear that all the axioms are met. For any number $a \in \mathbb{R}$, its additive inverse would be $-a$, and its multiplicative inverse would be a^{-1}. The identities represented by the symbols 0 and 1 above are in fact 0 and 1, and axioms 1, 2, 5, 6, 9, and 10 are obvious. The same is true for the rationals and the complex numbers.

1.1.5 $\mathbb{Z}$ is not a Field

Not every subset of a field is a field. The integers are not a field because it is not true that for any $a \in \mathbb{Z}$, $\exists b \in \mathbb{Z}$ with $a \times b = 1$. 2 is certainly an integer, but for $2 \times b = 1$ to hold, $b = \frac{1}{2}$, which is certainly not an integer!

1.1.6 The Binary Field

The set $\{0, 1\}$ with multiplication and addition as usual, except that $1 + 1 := 0$ is a field. It is the smallest field and the unique field with two elements. Addition defined in this way is known in computer science as XOR.

1.1.7 The Properties of Fields

From the axioms there is a standard set of proofs used to establish the properties of fields usually taken for granted.

1. Uniqueness of 0.

   The additive identity 0 is unique. Let 0 and $0'$ be additive identities, then $0+0'=0$ and $0'+0=0'$. From the commutativity of addition, $0+0'=0'+0$, and thus $0=0'$. $\square$

2. Uniqueness of 1.

   Let 1 and $1'$ be multiplicative identities, then $1\times 1'=1$ and $1'\times 1=1'$. But $1\times 1'=1'\times 1$, thus $1=1'$ $\square$

3. Uniqueness of the inverses.

   Let b and c be additive inverses of a, then

$$
\begin{aligned}
b &= b+0\\
b &= b+(a+c)\\
b &= (a+b)+c\\
b &= 0+c\\
b &= c
\end{aligned}
\qquad\square
$$

   Let b and c be multiplicative inverses of a, then

$$
\begin{aligned}
b &= b\times 1\\
b &= b\times(a\times c)\\
b &= (a\times b)\times c\\
b &= 1\times c\\
b &= c
\end{aligned}
\qquad\square
$$

4. Cancellation laws

   If $a+b=a+c$, then $b=c$, and if $a\neq 0$ and $ab=ac$ then $b=c$.

5. Arithmetic of 0.

   $a \times 0 = 0$

6. Sign rules

   $(-1) \cdot a = -a$ and $(-a) \cdot b = -(ab) = a \cdot (-b)$

7. No zero divisors

   If $ab = 0$ then $a = 0$ or $b = 0$. A zero divisor is a non zero element a, such that $\exists b \neq 0$ with $ab = 0$.

1.1.8 Power Sets

A POWER SET $P(S)$ is the set of all subsets of S. The set

$$S = \{1, 2, 3\}$$

has

$$P(S) = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}, \varnothing\}$$

If we define multiplication to be $A \cap B$ and addition to be the symmetric difference, $A \cup B \setminus A \cap B$, then all the axioms of a field are met *except* the existence of a multiplicative inverse. The multiplicative identity is the set $\{1, 2, 3\}$ since the intersection of any member of $P(S)$ with $\{1, 2, 3\}$ is itself, but it is not possible for every member of $P(S)$ to intercept with $\{1, 2, 3\}$ and produce $\{1, 2, 3\}$. A structure that satisfies every axiom of a field except the existence of a multiplicative inverse is called a *commutative unital ring*.

1.1.9 Boolean Rings

A RING is a structure consisting of a set with two operations $+$ and $\times$ which behave similarly to how multiplication and addition behave with integers, except that multiplication does not need to be commutative. For example, the set of all 2×2 matrices with real or complex entries, with standard matrix addition and multiplication, form a ring.

A COMMUTATIVE RING is a ring in which multiplication is commutative, i.e. $ab = ba$. The difference between a commutative ring and a field is that commutative rings do not necessitate the existence of multiplicative inverses or a multiplicative identity. A *unital* commutative ring does contain a multiplicative identity, hence its name. The set of all even integers, $2\mathbb{Z}$, with standard addition and multiplication form a commutative, but not unital, ring, since the identity 1 is not contained within the set. The non-existence of a multiplicative identity implies the non-existence of a multiplicative inverse. The set of all even integers, and 1, would be a unital commutative ring.

Idempotence is the property that repeated application of an operation does not alter the outcome after the first application. Repeatedly turning a light on whilst it is already on is an idempotent operation. In mathematics, an element is idempotent under $\times$ if $x \times x = x$. The operation $\times$ is itself idempotent if $x \times x = x \ \forall x \in S$. 0 and 1 are idempotent elements of the field of real numbers, and the absolute value function is an idempotent function, since $|(|f|)| = |f|$; infact, $|(|f|)|^n = |f|$.

If multiplication of two sets were to be defined as $\cap$, then $A \cap A = A$, and that would be the case no matter how many times you multiplied A with A.

A boolean ring is a ring in which every element is idempotent under $\times$. Idempotence implies commutativity, but a boolean ring does not necessarily contain a multiplicative identity. The set of all finite subsets of $\mathbb{N}$ with addition being symmetric difference and multiplication being intersection is a boolean ring that does not contain an identity, since for $A \cap E = A$, $E = \mathbb{N}$, which is not a finite subset of $\mathbb{N}$. The set described above is the power set of $\mathbb{N}$. The power set of any finite set is a unital boolean ring.

Chapter 2

Equivalence Relations

2.1 Lecture 2

2.1.1 Binary Relations

The cartesian product of two sets A and B is the set of all ordered pairs (a,b) such that $a \in A$ and $b \in B \rightarrow A \times B = \{(a,b) : a \in A,\ b \in B\}$.

A binary relation R is any subset of the cartesian product. For example, $R = \{(a,a) : a \in A\}$ is a subset of the cartesian product and is known as *equality*. If $(a,b) \in R$, then a is said to be *related* to b.

Choosing different subsets of $A \times B$, we get different relations, for example:

$$\begin{aligned} &A = \{1,2,3\} \quad B = A \\ &R = \{(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\} \end{aligned}$$

R is the relation, *less than or equal to*. The relation *divides*, for the sets

$$A = \{1,2,3,4\} \quad B = A$$

would be

$$R = \{(1,1),(1,2),(1,3),(1,4),(2,2),(2,4),(3,3),(4,4)\}$$

This represents what members of A divide members of B. Since 2 does not divide 3, it is not included in R. In this context, 2 is not related to 3. The sets A and B of the cartesian product $A \times B$, are called the domain and the codomain of the relation, respectively.

2.1.2 Relations Notation

INFIX NOTATION is the predominant notation for relations that have human readable names. Infix notation places the relational symbol between the two related objects. For example, $1 \leq 2$ or $2 \mid 4$. Is it just as correct to say $(1,2) \in \leq$ or $(2,4) \in \mid$. For more complicated relations, set notation is used. A rule to construct the relation, or a list of tuples for which the relation applies provides all the information required.

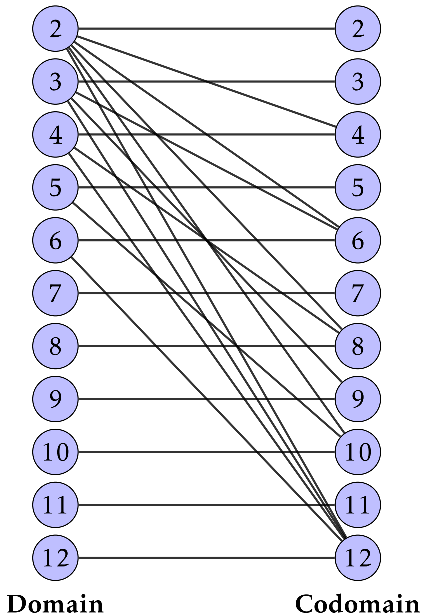

Figure 2.1: Bipartite graph representing the *divides* relation.

2.1.3 Bipartite Graphs

BIPARTITE GRAPHS are visual representations of relations. The domain and codomain are both represented, with and edge to show relatability. For the sets

$$A = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\} \quad B = A$$

The relation *divides* can be represented by fig. 2.1. Each edge represents divisibility. There is an edge joining 3 and 6 since $(3,6) \in |$, but no edge joining 3 and 7, since 3 does not divide 7. A bipartite graph can be drawn for any relation.

2.1.4 Functions are Relations

Functions are the special class of relations for which each member of the domain has exactly one tuple in the relation, also known as *many-to-one* relations. A function $f : \mathbb{R} \to \mathbb{R}$ is the relation

$$f = \{(x, f(x)) : x \in \mathbb{R}\}$$

A function is only defined if there is exactly one tuple in the relation for each *and every* member of the domain. A point in the domain that does not contain a tuple in the relation is called a discontinuity. The codomain of a function relation is also called the range. From fig. 2.2, $f(1) = 5$, $f(2) = 4$, $f(3) =$

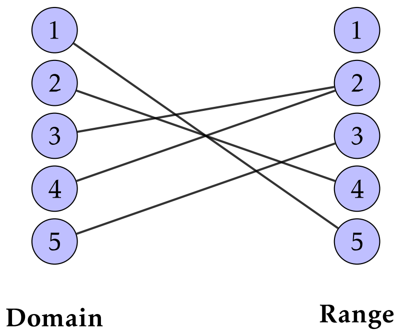

Figure 2.2: Bipartite graph of a function f

2, $f(4) = 2$, $f(5) = 3$. It is not necessary for each member of the range to only be represented once, or at all, just for each member of the domain to be represented exactly once.

2.1.5 Relations with $A = B$ and Directed Graphs

Relations with $A = B$ are said to be subsets of the cartesian product $A \times A$. R is a relation on A. A simpler graphical representation is available for relations where the domain equals the codomain.

Directed graphs contain a single representation of the domain and codomain with *directed* edges representing relatability. fig. 2.3 represents the relation of divisibility between $A = B = \{1, 2, 3, 4\}$. Small looping arrows are drawn for when an element relates to itself. Arrows can be left off if the relation is symmetric.

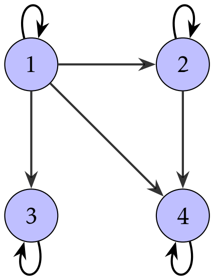

Figure 2.3: Directed graph of $R = \{(a, b) : a, b \in \{1, 2, 3, 4\} : a \mid b\}$.

For $A = 1, 2, 3, 4, 5$, $R = \{(a, b) : a, b \in A : a + 2b \leq 11\}$, two representations are possible. However, the bipartite fig. 2.4 contains a lot of redundant

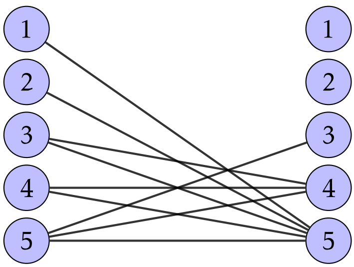

Figure 2.4: Bipartite graph of $R = \{(a, b) : a, b \in A : a + 2b \leq 11\}$

information, since $A = B$, so a directed graph (fig. 2.5) is preferred.

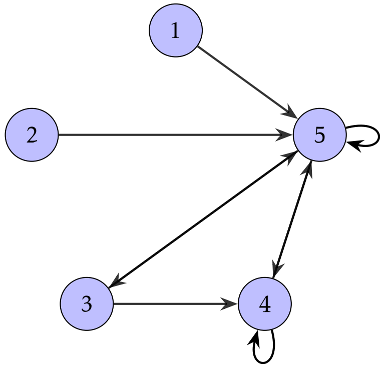

Figure 2.5: Directed graph of $R = \{(a,b) : a,b \in A : a + 2b \leq 11\}$.

2.1.6 Properties of Relations with Domain = Codomain

The relation is said to be:

1. **(R)** *Reflexive* if every element relates to itself, i.e. $(a,a) \in R,\ \forall a \in A$.
2. **(S)** *Symmetric* if $(b,a) \in R$ whenever $(a,b) \in R$.
3. **(T)** *Transitive* if $(a,c) \in R$ whenever $(a,b) \in R$ and $(b,c) \in R$.
4. **(A)** *Anti-symmetric* if $a = b$ when $(a,b) \in R$ and $(b,a) \in R$.

Reflexive relations contain every equality (a,a) for every $a \in A$. For example, the relation $\leq$ on the naturals (fig. 2.6).

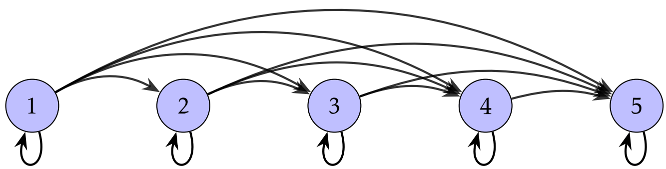

Figure 2.6: $R = \{(a,b) : a,b \in \mathbf{N} : a \leq b\}$. (Cut off at 5 for clarity).

Symmetric relations contain (b,a) whenever they also contains (a,b). Since addition is commutative, the relation $R = \{(a,b) : a,b,k \in \mathbb{N} : a+b = 2k\}$ is reflexive. Graphs (fig. 2.7) of reflexive relations contain no one-way edges.

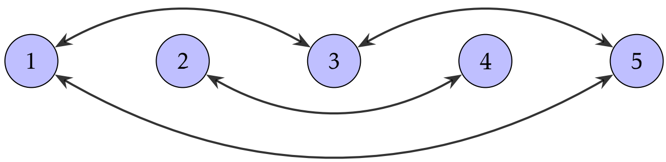

Figure 2.7: $R = \{(a,b) : a,b,k \in \mathbb{N} : a+b = 2k\}$. (Cut off at 5 for clarity).

Transitive relations contain (a,c) whenever they also contain (a,b) and (b,c). If you *walked* along a graph of a transitive relation, any journey that could take two steps could also take one. For example, $<$. (fig. 2.8)

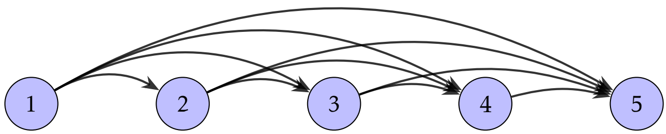

Figure 2.8: $R = \{(a,b) : a,b \in \mathbb{N} : a < b\}$. (Cut off at 5 for clarity).

2.1.7 Equivalence Relations

Equivalence relations are those that are reflexive, transitive, and symmetric. Consider $A = \{0,1,\ldots,8\}$ and $R = \{(a,b) : 3|a-b\}$. If we think of all the members of A related to 0, we get 0, 3, and 6. 0, 3, and 6 relate only to themselves and each other, and not to any other members of the domain. All possible relations of 0, 3, and 6 are represented by the first part of fig. 2.9. The same is true starting at 1 and 2. Each *directed clique* only relates to themselves and each other and never to other members of the domain. An equivalence relation is made up of a *disjoint union* of directed cliques. Each island of relatability does not interact with every other island. Equivalence relations are easily identifiable by their directed graphs, which always contain a disjoint unions of directed cliques, themselves containing every possible edge.

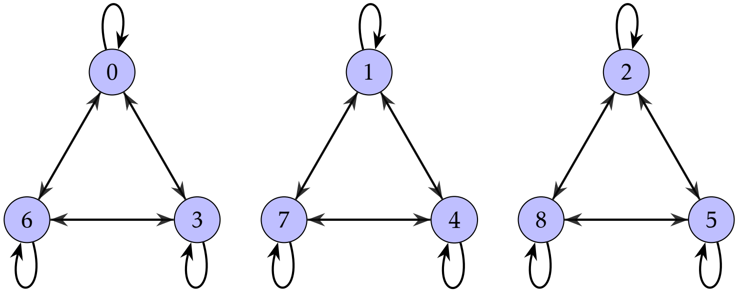

Figure 2.9: Directed cliques of R, forming a disjoint union of three components.

2.1.8 Equivalence Classes

For the general case R, being an equivalence relation on an arbitrary set A, define $[a]_R = \{b \in A : (a,b) \in R\}$. $[a]_R$ is the *equivalence class* of a and is all the elements that relate to a. It is a subset of A. The equivalence classes of 0, 1, 2 for the relation $R = \{(a,b) : 3|a-b\}$ are represented in fig. 2.9.

2.1.9 Properties of Equivalence Classes

1. $a \in [a]_R$ since it relates to itself (from reflexivity).
2. Any pairs $x, y \in [a]_R$ implies $(x,y) \in R$ (by construction).
3. If $b \in [a]_R$ then $[b]_R = [a]_R$ (from reflexivity, symmetry, and transitivity).
4. If $b \notin [a]_R$ then $[a]_R \cap [b]_R = \varnothing$ from 3.

2.1.10 Partitions

A partition of a set is any unique *region* of that set. Partitions do not share any elements but together they make up the whole set. Consider the set $X = \{0,1,2,3,4,5,6,7,8\}$ and $P = \{0,1,2 \mid 3,4 \mid 5,6,7 \mid 8\}$ where $|$ represents the partition border (fig. 2.10). If we define R on X to be any (x,y) in the same *cell* of P, then

$$R = \{(0,0),(0,1),(0,2),(1,0),(1,1),(1,2),(2,0),(2,1),(2,2),(3,3),(3,4),\\(4,3),(4,4),(5,5),(5,6),(5,7),(6,5),(6,6),(6,7),(7,5),(7,6),(7,7),(8,8)\}$$

which is a disjoint union of four equivalence classes (fig. 2.11). So, partition-

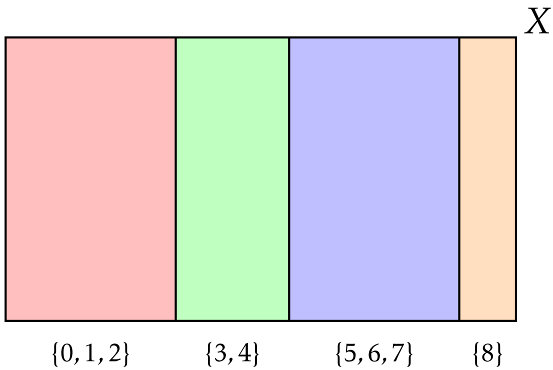

Figure 2.10: Partition diagram representing $P = \{0,1,2 \mid 3,4 \mid 5,6,7 \mid 8\}$.

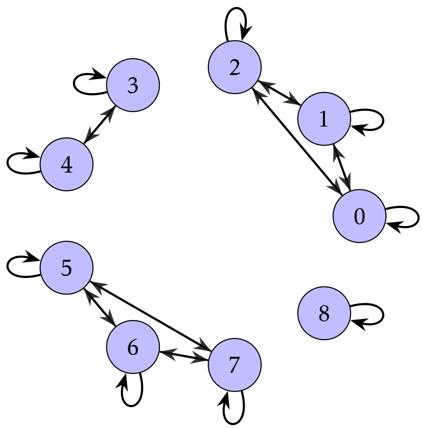

Figure 2.11: Directed graph of R.

ing sets gives rise to equivalence classes, and equivalence classes give rise to partitioned sets. Partitions and equivalence classes are two representations of the same underlying mathematical object.

2.1.11 Building the Rationals from the Integers

ASSUME we have integers, then the rationals are an equivalence relation on $\mathbb{Z}\times\mathbb{Z}\backslash\{0\}$. Specifically, for two tuples (a,b) and $(c,d) \in \mathbb{Q}$, $(a,b) \sim (c,d)$ if $ad = bc$, since $ad = bc$ is equivalent to $\frac{a}{b} \div \frac{c}{d} = 1$. For example, $(1,2) \sim (2,4) \sim (-30,-60)$ are all representations of the same number 0.5.

ANY RATIONAL NUMBER $\frac{a}{b}$ is the unique equivalence class $[(a,b)]_{\mathbb{Z}\times\mathbb{Z}\backslash\{0\}}$ where (a,b) is usually the number *in lowest terms*. For example,

$$[(1,2)] = \{(1,2),(2,4),(3,6),(4,8),\ldots\}$$

This solves the problem of uniqueness. Without representation of the rationals in this way, there is no unique characteristic that seperates $\frac{1}{2}$ from $\frac{2}{4}$, as they are both the number 0.5. However, there is a unique *equivalence class* representing 0.5 in all its forms.